

INDUSTRIAL PRODUCT DATASHEET

2.4GHz FPC Antenna A-02 for XIAO ESP32-S3

2.4GHz FPC Antenna A-02 for XIAO ESP32-S3 delivers 16dBi peak gain and flexible installation for industrial sensor networks, agricultural telemetry, drone swarm relays, and embedded IoT gateways.

SKU

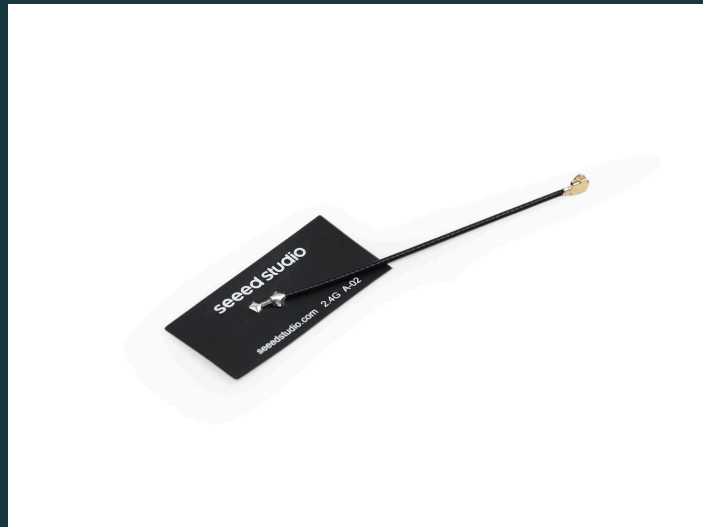
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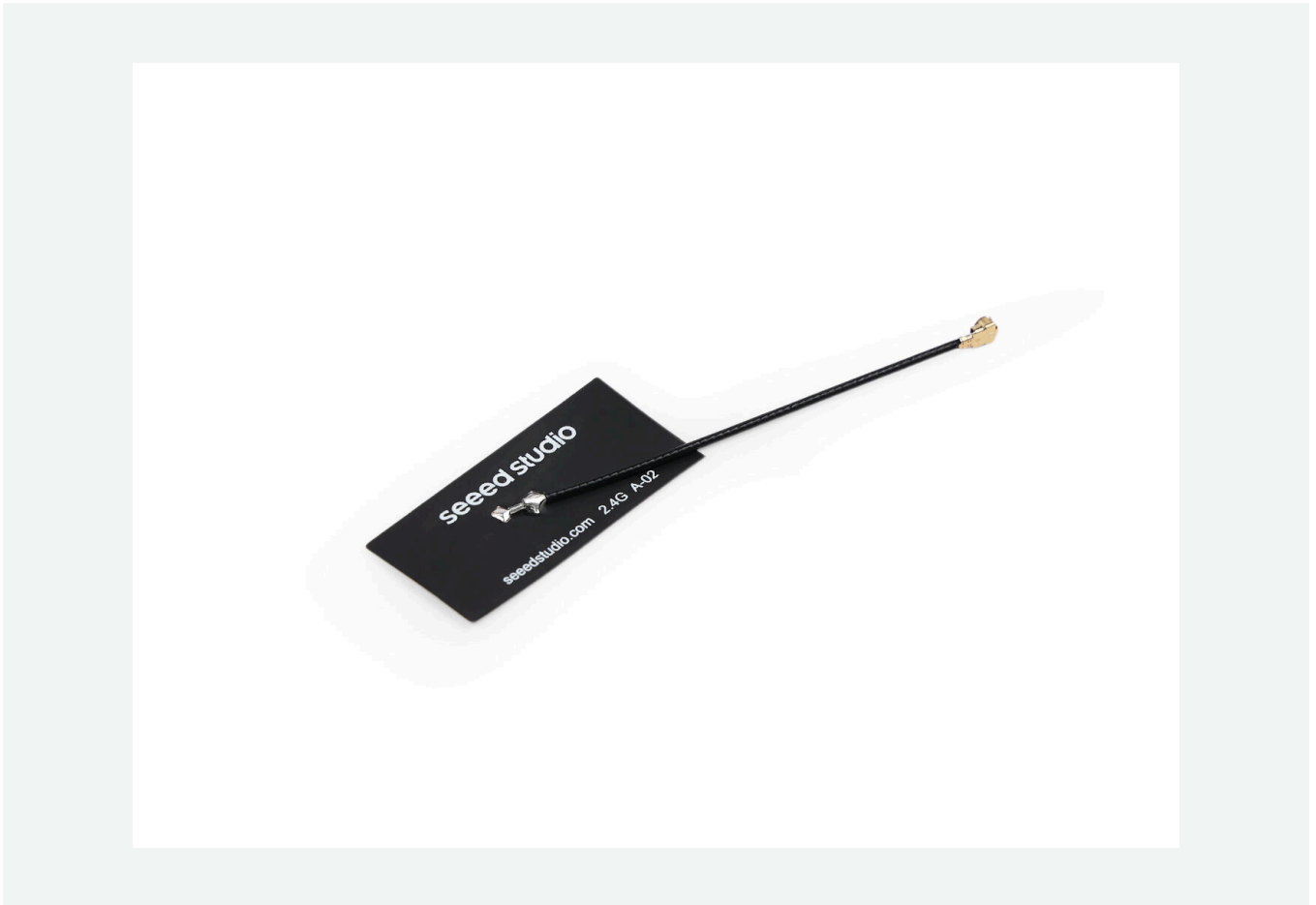
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PRICE

\$0.50(Excl. VAT)



Overview



This 2.4GHz FPC antenna enhances the wireless capability of Seeed Studio XIAO ESP32-S3 (Sense) boards. It features high-efficiency signal transmission with 16dBi peak gain and 50Ω impedance matching, and a flexible slim-profile FPC design for seamless integration in confined spaces.

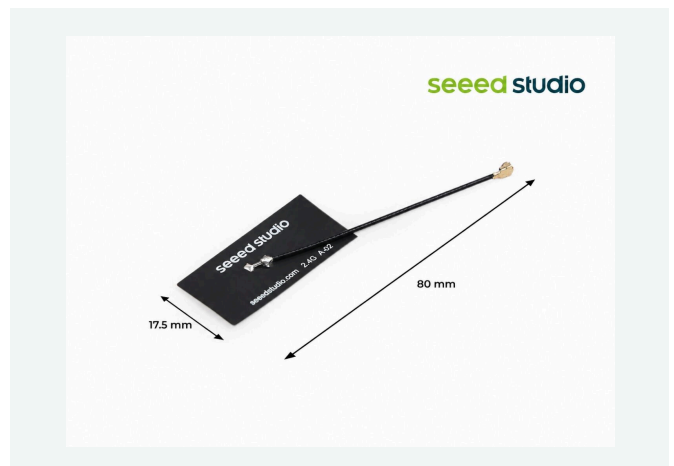
Key Features

Note:

Due to product updates, shipments may contain items with either the new or previous silk-screen markings. While the product's performance, quality standards, and functionality remain identical. You may use these products with full confidence.

- **Mini Antenna for Tiny Seeed Studio XIAO ESP32-S3 (Sense):** Attach everything you need to enhance the tiny MCU boards' wireless capability in 2.4GHz.
- **High-efficiency signal transmission:** 16dBi peak gain with 50Ω impedance matching ensures reliable long-range connectivity.
- **Flexible slim-profile design:** FPC material construction provides durable bending capability for seamless integration in confined spaces and complex installations.

Product Gallery

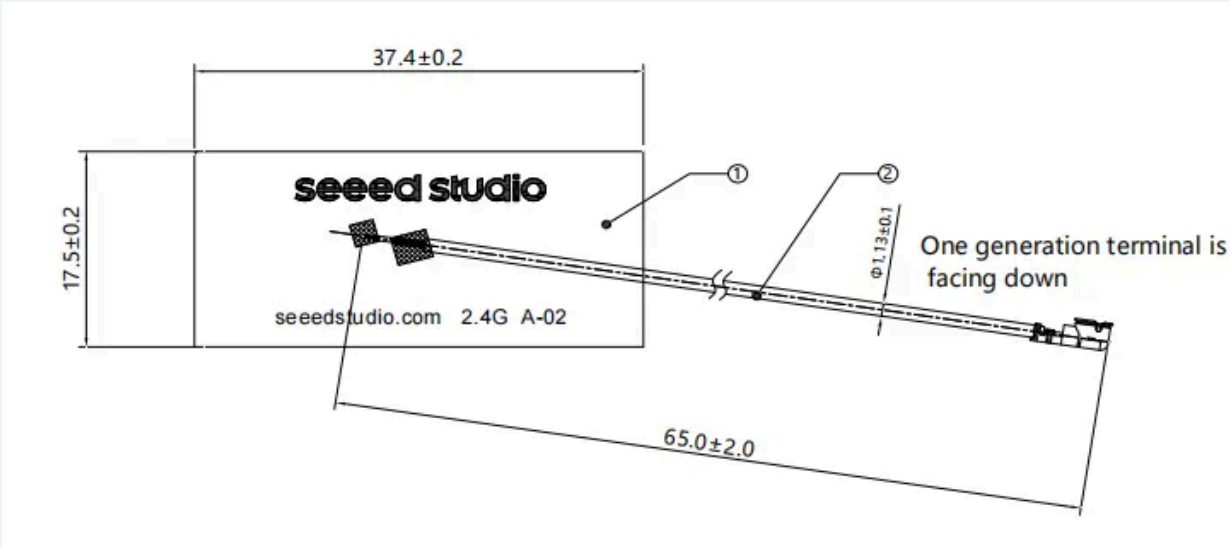


Specification

Parameter	Description
Frequency Bands	2400MHz~2500MHz
Peak Gain, dBi (Typ)	Up to 1.16
VSWR(Max)	4.5:1
Nominal Impedance	50Ω
Max Efficiency	61.84%
Operational Temperature Range	-40°C to +85°C
Storage Temperature Range	-40°C to +100°C
Polarization	Linear
Material Substance Compliance	REACH/RoHS Compliance

	2.4GHz FPC Antenna (1.16dBi) for XIAO ESP32-S3	2.4GHz /5GHz FPC Antenna for XIAO ESP32-C3/C5	2.4GHz Rod Antenna 2.81dBi for Seeed Studio XIAO
VSWR(Max)	4.5:1	3:1	3.0:1
Peak Gain, dBi (Typ)	1.16	2.8	2.81
Polarization	Linear	Linear	Linear, Omnidirectional
Compatibility	XIAO ESP32-S3(Sense)	XIAO ESP32-C3	XIAO ESP32-C3, XIAO ESP32-S3 (Sense)

Hardware Overview



Applications

- Industrial Sensor Networks
- Agricultural Telemetry Systems
- Drone Swarm Relays
- Embedded IoT Gateways

Part List

2.4GHz FPC Antenna (1.16dBi) for XIAO ESP32-S3	×1
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Compliance & Logistics

HSCODE	8517710000
USHSCODE	8517710000
UPC	
EUHSCODE	8531202090
COO	CHINA

Resource Links

- <https://www.seeedstudio.com/2-4GHz-FPC-Antenna-1-16dBi-for-XIAO-ESP32S3-p-6440.html>
- [Datasheet](#)

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